

Total No. of Questions : 8]

SEAT No. :

PE2472

[Total No. of Pages : 2

[6584]-451

**B.E. (Robotics) (Honors)**

**INDUSTRIAL ROBOTICS & AUTOMATION**

**(2019 Pattern) (Semester - VII) (404181)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.
- 4) Neat diagrams must be drawn whenever necessary.

- Q1)** a) Explain components and constructional features of Pneumatic control. [7]  
b) Explain pressure and flow of Pneumatic control. [7]  
c) What are the types of cylinders in Pneumatic control. [4]

OR

- Q2)** a) Draw and explain pneumatic control system design. [7]  
b) Explain in detail control modes, program control and sequence control in Pneumatic control system design. [7]  
c) Short note on Karnaugh – Veitch mapping. [4]

- Q3)** a) What are the special design features of CNC systems and features for lathes and machining centers? [7]  
b) What are condition monitoring of manufacturing systems? [6]  
c) Short note on CIM. [4]

OR

- Q4)** a) Design parts for high speed feeding and orienting. [7]  
b) Analysis of an assembly for programmable automation. [6]  
c) General rules for product design for programmable automation. [4]

- Q5)** a) What are the stages in design of mechatronics systems? [7]  
b) Draw and explain in detail pick and place robot. [7]  
c) Applications of mechatronics systems. [4]

OR

**P.T.O.**

- Q6)** a) What are possible design solutions for mechatronics systems? [7]  
b) Draw and explain in detail engine management system. [7]  
c) Compare traditional and mechatronics design. [4]

- Q7)** a) What are types of pumps and motors of hydraulic systems? [7]  
b) Selection criteria for cylinders, valves and pipes. [6]  
c) How heat generation in hydraulic system. [4]

OR

- Q8)** a) Explain Electro pneumatic in detail. [7]  
b) Short note on PLC. [6]  
c) Draw ladder diagram with suitable example. [4]

